Ahmed Yesuf **Nurye**

Ph.D. Candidate · Human-Interactive Robot Learning
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Research Interests

My research is centered around making robots easier for people to teach. I work toward this goal by: 1) enabling robots to learn from naturally expressed human feedback, which may combine different teaching signals (e.g., demonstrations, corrections, and evaluations) and span different levels of abstraction, from high-level planning to low-level motion execution; 2) accounting for differences in people's knowledge and preferences, so robots can adapt to individual users and learn from imperfect feedback; and 3) making the learning process more efficient by enabling robots to proactively participate in the teaching process and guide people toward providing feedback that is both useful for learning and easy to provide.

Education

Vrije Universiteit Amsterdam — *Amsterdam, The Netherlands*
Ph.D. in Computer Science — *Oct. 2025 – Oct. 2029*
- Advisors: Prof. Kim Baraka & Prof. Herke van Hoof (University of Amsterdam).

Warsaw University of Technology — *Warsaw, Poland*
M.Sc. in Robotics and Automatic Control — *Oct. 2022 – Oct. 2024*
- Advisor: Prof. dr hab.inż. Elżbieta Jarzębowska.
- Thesis: Deep Reinforcement Learning for Mobile Robot Navigation in Dynamic Environments.

Addis Ababa Science and Technology University — *Addis Ababa, Ethiopia*
B.Sc. in Electrical Engineering — *Oct. 2016 – Sep. 2021*

Experience

The Hybrid Intelligence Centre, VU Amsterdam — *Amsterdam, The Netherlands*
Doctoral Researcher — *Oct. 2025 – Present*
- Developing interactive robot learning methods that enable robots to learn from diverse human feedback.

Scania Group — *Gdańsk, Poland*
Systems Engineer — *Apr. 2024 – Oct. 2025*
- Developed safety-critical embedded software for automotive battery systems and contributed to its end-to-end verification and validation.

Northvolt — *Gdańsk, Poland*
Systems Engineer — *Apr. 2024 – Apr. 2025*
- Performed end-to-end V&V of battery management system functions and developed Python packages to automate code generation and streamline system integration.

Publications

🎓 Google Scholar; † → Equal contribution; ‡ → Equal supervision.

Conference Proceedings

[C2] **Show & Tell: Hierarchical Robot Learning with Human Corrective Feedback**
A. Y. Nurye, K. Baraka, and H. van Hoof
IEEE International Conference on Robotics and Automation, ICRA, 2027 (submitted)

[C1] **Deep Reinforcement Learning for Mobile Robot Navigation in Dynamic Environments**
A. Y. Nurye and E. Jarzębowska
2025 29th International Conference on Methods and Models in Automation and Robotics (MMAR), 2025

Teaching

Introduction to Control System (EEEg4155) — *Addis Ababa Sci. & Tech. University*
Teaching Assistant — *2022*

Electrical Measurement & Instrumentation (EEEg3153) — *Addis Ababa Sci. & Tech. University*
Teaching Assistant — *2021*

Skills

Programming Languages	Python, C/C++, MATLAB, Bash
Robotics & Simulation	ROS 2, MuJoCo, Gazebo, Isaac Lab, Simulink
Software & Infrastructure	Linux, Git, GitHub, Docker

Tools and Software

gym-turtlebot: a turtlebot4 gymnasium environment *website*

Python | ROS2 | Gazebo Sim *Mar. 2025 – Present*

- A minimal ROS2 and Gazebo simulation setup for TurtleBot4 to rapidly prototype DRL navigation agents using the Gymnasium API.

MBD Simulink *GitHub*

MATLAB | Simulink *Dec. 2024 – Present*

- Automates block insertion, naming, and connections in Simulink to streamline model-based design workflows.

Awards and Honors

2022 **Banach Scholarship**, Fully funded scholarship for master's studies in Poland. *NAWA*